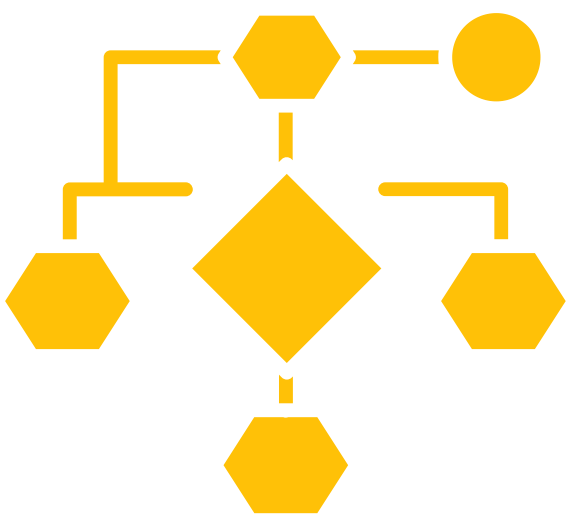
Introduction to AI Algorithms

What are Algorithms?

Algorithms are step-by-step instructions that tell computers how to process inputs to generate an output. This can be simple, fixed instructions, or it can be complex programs that adapt and learn from patterns. The latter is called machine learning, which is used to create Artificial Intelligence.



There are many different ways algorithsm are used with Artificial Intelligence.

Social Media Algorithms

Social Media platforms such as instagram or youtube uses AI algorithms to recommend videos to viewers. These algorithms are used to deliver content that is more relevant, useful or eye-catching to the user.



These platforms also use AI algorithms for larger content moderation. This enables the platform to filter out inappropriate, harmful or disrespectful content so that people are not exposed to such media. However, these come with their own set of challenges, such as moderation becoming censorship rather than a protective procedure.

Hiring Practices- Algorithms

Hiring practices and procedures have evolved greatly since the introduction of AI algorithms. Various companies(regardless of industry) have adopted AI hiring companions that select capable, and diverse hires. This increases the efficiency of the hiring process, but evokes new issues.



AI can reproduce existing human biases, meaning that it could result in a heightened discrimination in categories such as gender and race.

AI policing



One of the areas that AI excels at is data. Artificial Intelligence is good at absorbing and analyzing data. This is useful in policing, as it offers support in areas such as revealing trends in vast mounds of data, biometrics that allow quick identification of criminals.

However, this brings up challenges such as racial and gender profiling, resulting in minorities and specific demographics being targetted.

Privacy Concerns

One of the areas that AI excels at is data. Artificial Intelligence is good at absorbing and analyzing data. This is useful in policing, as it offers support in areas such as revealing trends in vast mounds of data, biometrics that allow quick identification of criminals.



However, this brings up challenges such as racial and gender profiling, resulting in minorities and specific demographics being targetted.

Social Media and AI Algorithms

Echo Chamber and Confirmation Bias



Social Media algorithms are designed to deliver information that is the most relevant and interesting to users. Though effective in capturing the audience's attention, this sort of filtering of information creates echo chambers of information.

Echo chambers are defined as environments in which specific opinions, perspectives or political leanings are reinforced due to continuous interaction with those with similar beliefs. Confirmation bias is caused by the tendency to seek out information that reinforces a specific belief, leading to users being exposed to content that generally agrees with what they believe.

These online echo chambers can cause these groups with similar opinons to move towards the extreme(polarization), as their exposure to information from the opposing side decreases significantly.

Echo chambers affect politics and misinformation as well. These echo chambers can interfere with the user's leaning for specific topics, creating extremism in certain beliefs. The exposure to evidence(true or otherwise) that only supports one claim causes bias towards the user's pre-existing belief.



Misinformation and Politics

Misinformation and Disinformation are two distinct phenomena. Misinformation is the spread of false information without the intention to be deceptive, while disinformation has the intent to deceive the audience.

The rise of deep fake technology and hallucinations from AI has enhanced both misinformation and disinformation in social media. Fake information is easier than ever to craft and to make it believable, and due to social media being opened to everyone, misinformation is prevelant in all areas of the internet.

Disinformation and misinformation affects politics in particular. The creation of false footage of speeches, diplomatic meetings etc instills misunderstandings and create political polarization.

"Ragebait" and Interaction



Social Media posts that drive powerful, intense emotions such as rage, joy or awe are much more likely to create interactions such as likes, comments or shares. These posts are then elevated due to social media platform algorithsm flagging them as being popular. This creates a bandwagon effect where these posts are much more likely to be shown in social media feeds.

This makes creators favor the creation of posts and videos that insight strong emotions rather than making high-quality, meaningful content.

Policing with Artificial Intelligence

Discrimination in Profiling

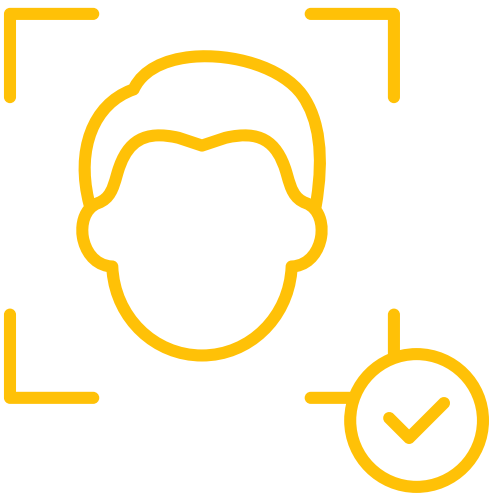
Artificial Intelligence has revolutionized criminal profiling, powering through traditional behavior analysis methods by collecting and processing vast mounds of data to detect patterns and decipher/predict criminal behavior. However, this does not come without its challenges with bias and other ethical considerations.



Predictive policing tools that use AI are much more vulnerable to biases such as racism and sexism. These tools make assessments of who is likely to commit future crimes and where these crimes will occur. Therefore they are are likely to enhance existing racial profiling issues, as police have historically focused on areas of certain ethnicity and therefore an overwhelming proportion of police data over-representing certain racial groups as being more likely to commit crimes.

This may cause a vicious cycle, where AI increases deployment into such over-represented areas, with the resulting police data being fed back to to the tool. Therefore, perpetuating a cycle of racial profiling.

Facial Recognition Software & Biometrics



The use of Artificial Intelligence Facial Recognition software has become increasingly popular in the past few years. The British police has adopted live facial recognition software, having doubled the number of faces scanned in the past year. However, this technology is not as effective as it claims to be with there being higher error rates with recognizing people of color.

The collection of biometric data, not just the face, has raised concerns of privacy as well. Fingerprints and face structures can not be rewritten or changed as easily as passwords, leading an increase in the risk that biometric data leaks pose. For example, there has been leaks from the US government such as in 2015 where the finger print data of approximately 5.6 million government employees were released.



Legal systems and AI



Artificial Intelligence can be vastly useful in the legal field. Both prosecutors and attorneys face the problem of an over-flood of information and data. AI systems can mitigate this issue, allowing individuals to comb through vast amounts of data and draw conclusions from it without becoming overwhelmed.

However, AI poses significant risks in ethical areas as well. Exposing AI to private and sensitive legal information might lead to data leaks and exposures, bringing up the question of whether AI can maintain legal confidentiality; experts question whether using AI is worth the risks. In addition, AI tools may make predictions, assessments, and judgements that are discriminatory against certain groups based on race or socioeconomic status.

Hiring Practices with AI

Hiring with AI

Artificial Intelligence tools can help automate parts of the hiring process for companies, such as filtering through hundereds of job applications to find applicants with a good fit to work at the company. However, the use of AI to automate reading the applications can cause systemic rejection of certain applicants based on personal data such as age, race and gender.



Job Applications with AI



The use of generative AI to write job applications have distorted the process of hiring. Unfortunately, this is not the only way job applicants use AI to gain an advantage when applying for jobs. Interviews have been considered the most authentic method to determine the genuinity or capability of a person. However, even interviews have been corrupted by AI systems that listen to the question and generate scripts on the go for interviewees to read during the interview. This is enhanced by the fact that many job interviews happen online since the start of the Covid-19 pandemic.

Privacy Concerns

As more and more companies have started using AI algorithms, there are also growing concerns of data privacy and personal privacy.



Companies have begun to collect sensitive data without permission, use data to train AI without consent, initiate unchecked surveillance of online activity etc.

More and more companies are seeking out data that they can use to train AI, and this comes with the fact that sensitive information such as biometrics, names, addresses etc can be given to AI for training.



This also inadvertently comes to the risk of data leaks. The sheer, vast amount of data used to train AI is susceptible to public exposure. Due to the scale of the data collected for AI training, it becomes a target for data miners and hackers. People can extract sensitive information from AI Agents through prompt injection, essentially disguising malicious input as legitimate prompts.

AI agents are vulnerable to accidental data breaches as well. Small AI tools used for medical purposes that contain sensitive patient data could be shared under specific prompts.